

6. Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(4n^2-1) \quad (6)$$



4. Prove that, for $n \in \mathbb{Z}$, $n \geq 0$

$$\sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)} = \frac{(n+a)(n+b)}{c(n+2)(n+3)}$$

where a , b and c are integers to be found.

(5)

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6. Prove by induction that for all positive integers n

$$f(n) = 3^{2n+4} - 2^{2n}$$

is divisible by 5

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6. (i) Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^n (3r+1)(r+2) = n(n+2)(n+3) \quad (6)$$

- (ii) Prove by induction that for all positive **odd** integers n

$$f(n) = 4^n + 5^n + 6^n$$

is divisible by 15

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4. (a) Use the method of differences to prove that for $n > 2$

$$\sum_{r=2}^n \ln\left(\frac{r+1}{r-1}\right) \equiv \ln\left(\frac{n(n+1)}{2}\right)$$

(4)

- (b) Hence find the exact value of

$$\sum_{r=51}^{100} \ln\left(\frac{r+1}{r-1}\right)^{35}$$

Give your answer in the form $a \ln\left(\frac{b}{c}\right)$ where a , b and c are integers to be determined.

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4. Prove by induction that for $n \in \mathbb{N}$

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix}$$

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4. Use the method of differences to show that

$$\sum_{r=1}^n \frac{2}{(r+4)(r+6)} = \frac{n(an+b)}{30(n+5)(n+6)}$$

where a and b are integers to be determined.

(6)



Answer ALL questions. Write your answers in the spaces provided.

1. (a) Prove that

$$\tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad -k < x < k$$

stating the value of the constant k .

(5)

- (b) Hence, or otherwise, solve the equation

$$2x = \tanh\left(\ln\sqrt{2-3x}\right)$$

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4. The infinite series C and S are defined by

$$C = \cos \theta + \frac{1}{2} \cos 5\theta + \frac{1}{4} \cos 9\theta + \frac{1}{8} \cos 13\theta + \dots$$

$$S = \sin \theta + \frac{1}{2} \sin 5\theta + \frac{1}{4} \sin 9\theta + \frac{1}{8} \sin 13\theta + \dots$$

Given that the series C and S are both convergent,

(a) show that

$$C + iS = \frac{2e^{i\theta}}{2 - e^{4i\theta}} \quad (4)$$

(b) Hence show that

$$S = \frac{4\sin \theta + 2\sin 3\theta}{5 - 4\cos 4\theta} \quad (4)$$

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4. (a) Use de Moivre's theorem to prove that

$$\sin 7\theta = 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta \quad (5)$$

- (b) Hence find the distinct roots of the equation

$$1 + 7x - 56x^3 + 112x^5 - 64x^7 = 0 \quad (5)$$

giving your answer to 3 decimal places where appropriate.

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5. (a)

$$y = \tan^{-1} x$$

Assuming the derivative of $\tan x$, prove that

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad (3)$$

$$f(x) = x \tan^{-1} 4x$$

(b) Show that

$$\int f(x) dx = Ax^2 \tan^{-1} 4x + Bx + C \tan^{-1} 4x + k$$

where k is an arbitrary constant and A , B and C are constants to be determined.

(5)

(c) Hence find, in exact form, the mean value of $f(x)$ over the interval $\left[0, \frac{\sqrt{3}}{4}\right]$ (2)

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2.

$$\mathbf{A} = \begin{pmatrix} 4 & -2 \\ 5 & 3 \end{pmatrix}$$

The matrix $\mathbf{A}$ represents the linear transformation M .

Prove that, for the linear transformation M , there are no invariant lines.

(5)

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3. $\mathbf{M} = \begin{pmatrix} 3 & a \\ 0 & 1 \end{pmatrix}$ where a is a constant

(a) Prove by mathematical induction that, for $n \in \mathbb{N}$

$$\mathbf{M}^n = \begin{pmatrix} 3^n & \frac{a}{2}(3^n - 1) \\ 0 & 1 \end{pmatrix} \quad (6)$$

Triangle T has vertices A , B and C .

Triangle T is transformed to triangle T' by the transformation represented by $\mathbf{M}^n$ where $n \in \mathbb{N}$

Given that

- triangle T has an area of 5 cm^2
- triangle T' has an area of 1215 cm^2
- vertex $A(2, -2)$ is transformed to vertex $A'(123, -2)$

(b) determine

- the value of n
- the value of a

(5)

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5. (a) Given that

$$y = \arcsin x \quad -1 \leq x \leq 1$$

show that

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \quad (3)$$

(b) $f(x) = \arcsin(e^x)$ $x \leq 0$

Prove that $f(x)$ has no stationary points.

(3)

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6. Given that

$$y = e^{2x} \sinh x$$

prove by induction that for $n \in \mathbb{N}$

$$\frac{d^n y}{dx^n} = e^{2x} \left(\frac{3^n + 1}{2} \sinh x + \frac{3^n - 1}{2} \cosh x \right)$$

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